

Supplementary Material S1

Single-cell and T-cell receptor analyses

This supplement reports two analyses performed alongside the reproducibility study in the main text. They address a different question — mechanisms of resistance and the cellular composition of the microenvironment — and they are reported separately because they do not bear on whether the two published signatures reproduce. Every number below is read directly from the result files in the public repository; the scripts that produced those files are named in each subsection.

The two components differ in scale and must not be described together as one longitudinal cohort. The single-cell component covers 12 patients with one sample each. The T-cell receptor component covers 42 libraries drawn from 34 patients, of whom only 7 contributed more than one biopsy; it therefore supports a cross-sectional comparison with a small paired subset, and no within-patient trajectory analysis is reported.

S1.1 Single-cell component: data and quality control

Source series GSE294273 (Nat. Commun. 2026, 17, 7445; main text reference 7). Script: `scripts/05–scripts/07, scripts/13, scripts/21, scripts/22`.

Of 88,715 nuclei passing initial loading, 85,167 (96.0%) survived quality control, resolving into 41 clusters. Per-sample quality-control metrics are in `qc/scrna_qc_per_sample.tsv` and are reproduced below.

Sample	Patient	Treatment	Nuclei after QC	Median genes	Median UMI	Median MT%	MT% p95
GSM8901614	HM002	Untreated	9,284	1346.0	2549.5	4.12	14.27
GSM8901615	HM004	ICI-resistant	7,475	1762.0	4474.0	3.08	8.83
GSM8901616	HM005	Untreated	8,919	1581.0	3829.0	2.92	9.71
GSM8901617	HM006	ICI-resistant	10,253	2003.0	5956.0	2.58	7.68
GSM8901618	HM009	ICI-responder	6,502	1792.0	4211.0	2.43	11.19
GSM8901619	HM012	Untreated	8,714	1631.0	3810.5	3.26	15.01
GSM8901620	HM014	Untreated	6,323	1659.0	4117.0	3.53	9.25
GSM8901621	HM019	ICI-resistant	1,749	1929.0	5687.0	2.84	9.85
GSM8901622	HM026	ICI-responder	3,960	1760.0	4581.0	2.77	8.0
GSM8901623	HM028	ICI-resistant	10,376	2535.0	7894.5	1.47	6.04
GSM8901624	HM035	ICI-responder	6,620	1963.0	5371.5	1.9	7.88
GSM8901625	HM038	ICI-responder	4,992	1700.5	5209.5	2.69	8.97

S1.2 Cluster markers

Top-30 differentially expressed genes per cluster by Wilcoxon rank-sum test, from `qc/scrna_cluster_markers.tsv` (1,230 rows). The top 5 genes of each cluster are listed; the complete table is in the repository.

Cluster	Top 5 markers
0	CTLA4, IL32, TIGIT, B2M, CD2
1	RPL28, RPS12, RPL10, RPL11, RPL13
2	JUNB, CD69, ZFP36L2, LTB, TXNIP
3	IL7R, SRGN, SARAF, CCR7, ANXA1
4	RPL34, RPS3A, RPS27A, RPLP2, RPS14
5	MALAT1, MT-CO1, MT-CO2, PTPRC, TNFAIP3
6	MALAT1, FYN, AC016831.7, IL7R, HSPH1
7	NKG7, CCL5, CST7, GZMA, LAG3
8	CD74, HLA-DPB1, HLA-DPA1, LSP1, HLA-DRA
9	CXCL13, CD7, CCL5, HLA-C, UBC
10	CCL5, NKG7, CD8A, CST7, LAG3
11	CCL5, GZMK, CST7, DUSP2, ZFP36L2
12	SERPINE2, MT2A, APOD, FTL, VIM
13	THBS2, FKBP10, RCN3, ITGB8, CALU
14	MT-ATP6, MT2A, MT-CYB, MT-ND4L, MT-ND2
15	FTL, FTH1, C1QB, C1QA, C1QC
16	TYROBP, PSAP, PLAUR, SOD2, FCER1G
17	IL3RA, IRF4, SLC15A4, IRF8, IRF7
18	CD79A, CD74, HLA-DRA, HLA-DPA1, EZR

Cluster	Top 5 markers
19	MS4A1, CD83, MALAT1, BANK1, AC016831.7
20	GAPDH, VIM, ACTB, PPIA, CD63
21	MS4A1, CD79A, BANK1, CD83, CD37
22	SARAF, RPL13, RPS27A, RPS26, RPS12
23	CTSW, GNLY, AREG, KLRD1, TYROBP
24	HMGB2, MKI67, STMN1, TUBA1B, TMPO
25	POU2AF1, CD79A, MZB1, SEL1L3, UBE2J1
26	CD79A, MS4A1, CD74, BANK1, HLA-DRA
27	CD79A, CD37, CD74, MS4A1, HLA-DRA
28	IGKV3-20, IGHV4-34, CD79A, CD37, MS4A1
29	IGFBP7, RAMP2, PECAM1, CRIP2, VWF
30	TRIM33, MDM4, EDNRB, MBNL2, EPS8
31	MALAT1, PDE3B, PTPRC, XIST, BACH2
32	CD8B, RPS3A, RPL34, RPS12, EEF1A1
33	CA8, MCAM, L1CAM, FN1, PMEL
34	IGFBP2, AKAP12, POSTN, AEBP1, STXBP1
35	SEMA5A, GDF15, MCAM, AEBP1, AKAP12
36	MGP, IFITM3, HSPA1A, SNRPE, CDK4
37	A2M, SFRP1, COL11A1, S100A1, ISYNA1
38	GAPDH, S100A1, RPL8, LGALS1, IFI27
39	MFSD12, PMEL, PAEP, LY6K, GNG11
40	MGP, BANC1, PAGE5, SNRPE, BCAS2

S1.3 The pre-specified tie-break rule and its application

The tie-break rule was written into the pre-registration before any tie-break result existed. Clustering and the marker table are reused unchanged from the locked run; the resolution was not revisited. For each cluster the top-30 markers are tested for hypergeometric enrichment within each candidate reference profile, using the full set of detected genes as the population and $\alpha = 0.05$; a cluster that fails to reach significance for any candidate is labelled `Unassigned` rather than being forced into the nearest profile.

Applied to 41 clusters, the rule changed the assignment of **11** clusters and left **6** clusters unassigned, together **15.1%** of all nuclei.

S1.3.1 Clusters whose assignment changed

Cluster	Original	New	Best candidate	Markers hit	Hypergeometric p
4	T_CD4	Unassigned	T_CD4	0/30	1.00e+00
5	T_CD8	Unassigned	T_CD8	0/30	1.00e+00
9	T_CD8	NK	NK	2/30	1.62e-05
13	Melanocyte	Unassigned	Melanocyte	0/30	1.00e+00
14	NK	Melanocyte	Melanocyte	1/30	6.25e-03
16	Macrophage	DC	DC	2/30	1.62e-05
24	T_CD8	Unassigned	T_CD8	0/30	1.00e+00
25	Plasma	B	B	4/30	1.82e-11
34	Melanocyte	Unassigned	Melanocyte	0/30	1.00e+00
37	Melanocyte	Fibroblast	Fibroblast	1/30	6.25e-03
38	Melanocyte	Unassigned	Melanocyte	0/30	1.00e+00

S1.3.2 The unassigned clusters

Abstention here does not mean the cluster lacks biological meaning; it means the marker set available does not reach it. This is the only evidence a reader needs to judge whether the marker set should be extended, and it is given deliberately.

Cluster	Previously assigned	Top 5 markers at stake
4	T_CD4	RPL34, RPS3A, RPS27A, RPLP2, RPS14
5	T_CD8	MALAT1, MT-CO1, MT-CO2, PTPRC, TNFAIP3
13	Melanocyte	THBS2, FKBP10, RCN3, ITGB8, CALU
24	T_CD8	HMGB2, MKI67, STMN1, TUBA1B, TMPO
34	Melanocyte	IGFBP2, AKAP12, POSTN, AEBP1, STXBP1
38	Melanocyte	GAPDH, S100A1, RPL8, LGALS1, IFI27

One implementation correction is recorded here rather than repaired silently: the first version of the tie-break script set the hypergeometric population to the genes appearing within the top-30 lists rather than to the full set of detected genes. That population had itself been enriched by differential-expression filtering, which artificially suppressed enrichment and caused roughly forty per cent of cells to abstain. The population was restored to the full detected gene set. This was a bug fix, not a change of rule made after seeing results: the rule itself was not altered.

S1.4 Cell-type composition before and after the tie-break

From qc/scrna_reference_profile_v2.tsv. ‘Before’ is the first-pass assignment, ‘after’ the pre-specified tie-break.

Cell type	Before	After	Change
T_CD4	26.1%	21.5%	-4.6 pp
Melanocyte	25.7%	18.9%	-6.8 pp
Unassigned	0.0%	15.1%	+15.1 pp
T_CD8	20.9%	13.2%	-7.7 pp
B	11.4%	12.3%	+1.0 pp
T_reg	6.0%	6.0%	+0.0 pp
DC	0.3%	4.3%	+4.0 pp
Fibroblast	0.0%	3.8%	+3.8 pp
NK	2.3%	2.4%	+0.1 pp
Macrophage	5.4%	1.4%	-4.0 pp
pDC	0.8%	0.8%	+0.0 pp
Endothelial	0.2%	0.2%	+0.0 pp
Plasma	1.0%	0.0%	-1.0 pp

S1.5 T-cell receptor clonality

Source series GSE308435 (main text reference 8). Script: scripts/08. 42 libraries, all reported; none was excluded.

Median clonality is 0.031 (range 0.000–0.281). Per-library values are in qc/GSE308435_tcr_clonality.tsv.

The series is described upstream as 42 sequential biopsies from 34 patients. That count is a tally of libraries, not a paired design: only 7 of the 34 patients contributed more than one biopsy, and a single untreated anchor exists across the whole cohort. Any statement about within-patient change therefore rests on those few patients and none is made here.

S1.6 Cell-state scores

Per-sample dispersion of five immune state programmes across 42 samples, from qc/GSE308434_state_scores.tsv. Each programme is summarised by its standard deviation across nuclei, its 90th-minus-10th percentile range, and the percentage of nuclei in the upper decile.

Sample	Nuclei		antigen	ifng	res	proliferation	hypoxia	inf	nfkb	na_res	defence	stability	apoptosis	phospho	torc1	pi3k	apoptosis	siogenesis	apoptosis
	af-ter	QC kept																	
GSM9245398_F0118pre	24.5	11.1	12.3	11.1	8.6	25.2	17.2	16.8	12.5	28.2	18.8	27.0	26.4	25.7	25.8	24.0	15.4	25.6	25.6
GSM9245398_F0126pre	23.1	12.3	16.6	29.4	22.6	24.0	19.9	31.1	26.0	29.0	26.7	29.0	29.7	29.5	21.9	29.4	29.4	29.4	29.4
GSM9245398_F0210pre	20.1	17.2	12.1	26.5	22.0	22.9	15.1	31.9	28.8	30.8	29.8	30.9	34.4	28.8	13.0	29.5	29.5	29.5	29.5
GSM9245398_F0228pre	16.3	15.7	10.3	24.7	20.5	21.8	11.0	29.0	23.5	28.8	26.9	29.5	30.6	26.7	12.8	29.0	29.0	29.0	29.0
GSM9245398_F0313post	22.5	17.2	12.6	32.3	28.6	21.6	22.1	32.9	34.3	35.0	29.1	33.2	36.0	32.4	18.0	34.4	34.4	34.4	34.4
GSM9245398_F0319post	21.0	13.2	9.7	26.1	21.4	23.1	16.9	27.1	26.0	29.7	29.7	29.3	30.8	26.9	17.0	29.6	29.6	29.6	29.6
GSM9245398_F0415pre	21.1	27.5	9.0	27.7	20.8	22.7	17.4	28.6	27.7	29.9	27.0	29.6	31.1	28.2	16.0	28.8	28.8	28.8	28.8
GSM9245398_F0518pre	20.5	26.1	13.9	29.2	24.2	22.6	20.0	31.9	30.1	32.5	29.8	32.0	33.3	31.2	17.2	30.4	30.4	30.4	30.4
GSM9245398_F0625post	22.8	27.6	30.8	30.1	30.5	26.9	32.2	30.6	32.0	32.9	32.2	32.6	31.6	26.9	31.1	31.1	31.1	31.1	31.1
GSM9245398_F0729post	21.6	18.4	23.7	24.8	26.7	27.1	27.9	29.5	26.1	25.7	25.6	30.7	27.1	22.2	26.6	26.6	26.6	26.6	26.6
GSM9245398_F0816post	23.7	16.9	14.5	31.6	31.3	24.3	20.9	32.0	31.8	32.7	31.8	32.6	32.8	32.2	19.9	32.3	32.3	32.3	32.3
GSM9245398_F0914post	21.0	24.4	18.4	28.5	23.5	21.4	16.3	28.6	26.1	29.2	27.0	29.0	29.8	28.0	22.4	29.1	29.1	29.1	29.1
GSM9245398_F1025post	27.3	17.4	29.2	27.8	24.8	26.5	25.3	30.4	29.1	28.1	25.8	25.0	29.7	28.6	27.3	26.5	26.5	26.5	26.5
GSM9245398_F1213post	24.8	24.6	25.4	30.5	23.8	25.8	25.6	34.5	33.7	33.9	30.8	32.9	36.2	33.8	24.1	34.2	34.2	34.2	34.2
GSM9245398_F1228pre	26.1	19.1	18.7	26.1	24.4	24.8	24.8	25.7	31.0	30.3	27.2	26.9	34.2	28.4	21.4	30.3	30.3	30.3	30.3
GSM9245398_F1519pre	21.1	13.2	12.1	29.4	24.8	22.2	20.0	29.3	27.6	31.5	32.5	32.4	32.2	30.5	17.3	32.0	32.0	32.0	32.0
GSM9245398_F1610post	21.6	23.0	27.5	24.6	26.0	19.5	31.7	28.6	30.9	27.1	29.2	31.2	29.3	24.1	31.1	31.1	31.1	31.1	31.1
GSM9245398_F1618pre	20.0	21.2	18.0	25.3	20.0	20.0	18.2	29.9	25.9	30.5	30.1	30.0	32.8	28.1	18.3	29.2	29.2	29.2	29.2
GSM9245398_F1719post	21.8	17.2	13.3	30.6	32.8	21.2	19.3	31.6	31.3	30.3	31.1	33.6	32.7	33.3	18.2	32.6	32.6	32.6	32.6
GSM9245398_F1708post	33.5	34.8	33.2	35.4	34.6	33.2	32.7	33.3	34.7	34.7	34.7	35.5	34.9	34.3	28.7	34.4	34.4	34.4	34.4
GSM9245398_F2015post	21.5	13.1	35.8	25.1	23.4	17.4	34.8	33.1	36.6	35.3	35.6	36.9	35.3	20.3	37.5	37.5	37.5	37.5	37.5

Sample	Nuclei after QC	QC kept %	antigen	proliferation	hypoxia	inf_nfln	response	death	stage	colysis	phospho	torc1	pi3k	apoptosis	apoptosis	apoptosis	apoptosis	apoptosis
GSM92382199F2216	post	20.9	17.8	23.8	28.7	23.4	23.2	22.7	30.9	28.7	29.7	27.5	28.2	31.9	28.4	27.5	29.2	29.2
GSM92382199F2216	post	18.7	28.4	22.5	29.2	23.1	21.8	18.3	33.4	29.5	32.9	28.7	31.5	33.5	28.0	20.8	29.8	29.8
GSM92454298F2529	post	28.3	27.9	11.3	22.8	20.4	28.0	22.6	24.9	27.0	27.5	29.1	30.8	32.5	27.1	17.3	27.0	27.0
GSM92454298F2529	post	28.2	18.5	16.5	28.4	28.6	26.3	25.2	26.8	31.0	29.0	25.4	29.0	32.2	29.4	25.6	27.2	27.2
GSM92454298F2529	post	15.8	24.6	27.4	19.2	24.2	17.0	33.1	27.7	31.2	27.5	31.2	34.4	29.2	21.6	29.8	29.8	29.8
GSM92454298F2529	post	17.3	9.1	31.1	26.1	24.9	17.2	31.0	28.5	30.8	29.7	30.9	32.0	30.1	19.7	30.1	30.1	30.1
GSM92454298F2529	post	22.8	12.7	33.0	29.9	26.6	22.9	33.0	31.4	32.8	29.4	32.4	34.5	32.4	18.3	32.3	32.3	32.3
GSM92454298F2529	post	25.4	20.1	22.4	30.7	26.3	27.0	25.7	32.8	30.9	31.9	29.0	30.7	33.6	30.3	24.7	31.9	31.9
GSM92454298F2529	post	31.1	27.7	28.9	32.6	30.9	31.8	29.3	32.9	33.0	32.2	32.1	32.3	33.7	32.7	27.2	31.9	31.9
GSM92454298F2529	post	24.0	13.1	14.5	31.1	28.2	26.7	22.8	35.2	31.4	32.9	33.6	31.8	34.3	33.2	16.1	32.9	32.9
GSM92454298F2529	post	24.2	15.4	18.4	29.9	23.5	25.2	20.9	32.4	30.7	32.8	31.5	31.8	33.7	32.2	16.3	30.9	30.9
GSM92454298F2529	post	26.0	19.0	20.2	28.0	24.3	26.2	24.7	29.9	31.8	26.2	23.7	25.7	32.1	30.2	20.7	28.9	28.9
GSM92454298F2529	post	12.1	13.7	9.8	28.4	17.7	23.1	17.1	31.3	27.2	33.0	28.2	31.7	34.4	28.1	17.1	30.9	30.9
GSM92454298F2529	post	22.3	17.7	17.5	27.1	25.1	25.9	20.3	29.3	31.0	26.5	25.7	27.7	31.9	30.0	20.6	28.4	28.4
GSM92454298F2529	post	20.2	15.4	8.8	27.2	19.1	25.6	15.9	29.5	26.9	30.6	29.9	30.9	31.4	27.3	16.2	30.1	30.1
GSM92454298F2529	post	21.9	15.6	11.0	25.7	20.6	23.3	17.1	27.9	25.8	29.6	28.2	29.9	30.5	28.0	17.6	29.8	29.8
GSM92454298F2529	post	22.5	18.7	15.1	25.7	20.2	23.5	19.2	28.1	25.4	27.9	26.6	26.8	28.8	27.6	19.8	28.4	28.4
GSM92454298F2529	post	18.3	16.5	11.6	27.5	19.5	20.9	16.3	27.6	25.6	29.4	28.7	28.2	29.7	27.2	17.1	27.1	27.1
GSM92454298F2529	post	26.8	18.8	22.7	29.9	27.6	27.4	25.6	30.6	28.7	29.2	28.7	28.4	31.4	30.0	21.2	29.9	29.9
GSM92454298F2529	post	20.3	20.4	14.9	28.7	25.6	20.8	20.2	32.8	31.4	32.1	30.6	31.8	33.8	31.1	17.3	32.0	32.0
GSM92454298F2529	post	20.4	26.8	11.1	26.4	23.4	23.1	19.8	30.6	28.8	31.4	28.3	30.3	31.8	29.4	18.9	29.7	29.7

S1.7 What this supplement does and does not support

Because a substantial fraction of nuclei could not be assigned with confidence, the deconvolution performed here supports coarse statements only — the broad division of the microenvironment into lymphoid, myeloid, melanocytic and stromal compartments. No claim of fine-grained cell-type decomposition is made, and none of these results is used to support or qualify the reproducibility conclusion in the main text.

All result files are in the public repository at the address given in the Data Availability statement, together with the scripts that produced them.